

Liquidity Providers in “Extreme” Periods: High Frequency Machines vs. Human Electronic and Floor Traders

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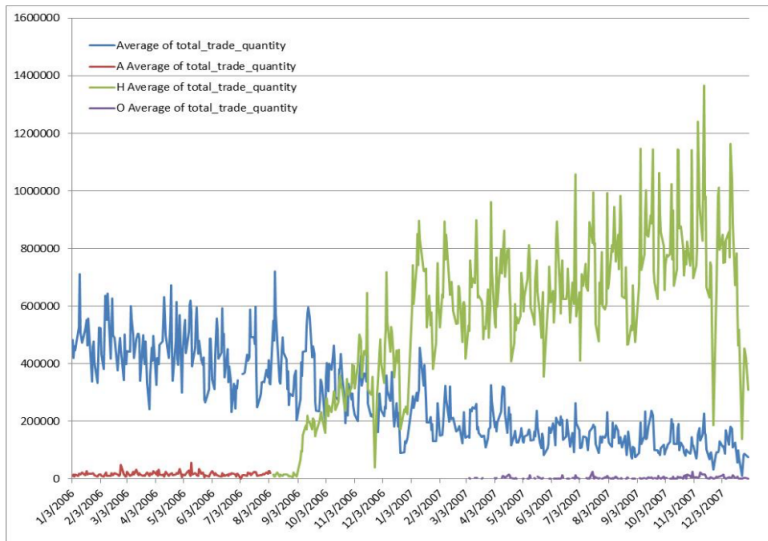
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Figure 1: Globex vs. Pit Trading Volumes, January 2006 to December 2007



Summary

- ▶ We care about market quality - ex-ante characteristics of market like liquidity
- ▶ Market participants (liquidity providers) are changing due to technology
- ▶ Participants' reliability may be conditional on the state of the market
- ▶ Do these changes matter for customers who demand liquidity?

Data

All intraday transaction records, NYMEX WTI light sweet crude oil futures and options on futures January 3rd 2006-June 30th 2009

- ▶ Allows customer-anchored view of liquidity
- ▶ Order imbalances and transaction costs of customers!
- ▶ Tagging of individual traders - focus on market-making HFT rather than all HFT
- ▶ Key - allows for tests of comparative behavior e.g. since the relative dominance of groups are changing we care about relative behavior

Key Concepts

Types of traders and their characteristics:

Locals

- ▶ Physically present in the trading pits
- ▶ Never trade more than 450 times a day
- ▶ Neither anonymous nor automated

e-Locals

- ▶ Trade electronically on Globex
- ▶ Trade less than 450 times a day
- ▶ Anonymous but not automated

HFT

- ▶ Anonymous and Automated
- ▶ Classified based on trade frequency - 6 or more trades per minute during trading hours for every day when the trader is active

Key Concepts : “Extreme Periods”

Key assumption:

Automated trading systems work well during market environments that can be modeled probabilistically but not in periods of “extreme uncertainty” (Knightian uncertainty, unknown unknowns)

Authors focus on two types of “extreme periods”:

- ▶ The minutes after one-hour episodes of persistently high minute-by-minute volatility (2sd above mean for 60mins),
- ▶ WTI inventory uncertainty.

Cushing Oklahoma “Extreme Periods”



- ▶ Authors use Fall 2008 Winter 2009 shortage of storage capacity
- ▶ 2.1 million barrels a week can go in, total storage around 71mil
- ▶ Prices depend heavily on inventory near the boundaries

Key Findings

1. HFT participate less and provide less liquidity in extreme periods
2. HFT react differently to extreme information driven periods as opposed to extreme periods driven by uninformed customer demand flow
3. HFT behavior differs from the behavior of e-Locals and Locals
 - ▶ Not driven by anonymity because results go through with HFT relative to e-Locals

Comment # 1 - Classification of HFT and Key Finding

“General regulatory concern of whether voluntary ELP reliably provide liquidity in unfavorable circumstances”

- ▶ Participation results in Table 7 - relative to e-Locals, HFTs have a lower volume share and provide less liquidity
- ▶ Tables 10,11: In normal periods HFT provide liquidity to customers at a lower cost than human liquidity providers but customer-human intermediary spreads don't change during extreme periods, but customer-HFT spreads do.

These results suggest HFT participate less in bad times, do they withdraw or transform their trading style?

Comment # 1 - Classification of HFT and Key Finding

Whether or not you are classified as HFT depends on rolling classification

- ▶ A HFT today may not be a HFT tomorrow - monthly rolling classification based on an average of 6 or more trades per minute during trading hours for each day for which they are active,
- ▶ Think about the same account, e.g. at Citadel
- ▶ Are we worried that automated makers don't have other skills? While I agree that they can't automate for every eventuality, maybe they are flexible and can "become" e-Locals in bad times...
- ▶ Is it possible that HFT pricing in good times pushes out locals, and then in bad times they can't provide liquidity in the same way?

Comment # 2 - Pure Liquidity Shocks vs. Information Shocks

Customer Demand Imbalance Variable is uninformative and used to measure “pure liquidity” shocks. HFT do not pull back from these pure liquidity shocks and there is no change in customer-HFT spreads during these periods...

- ▶ If this data is unobservable then it's not surprising their behavior doesn't change conditional on this data
- ▶ Otherwise, what is the mechanism through which they figure this out?
- ▶ Does this result go through with a more observable measure of uninformed order flow?

Comment # 3 - Profits and Strategies

The data seems to allow you to calculate MM profits:

1. Are e-locals doing worse when HFT come in?
2. What are different strategies? is it merely speed or something more fundamental?
3. Table 6 - locals' strategies are insensitive to market conditions... this is fascinating, all the benefit of HFT comes from "normal times"
4. Are locals bidding correctly? it could be that they get taken advantage of in extreme periods

Other comments

1. In general it would be interesting to see if customer-anchored views of liquidity are very different from the standard measures...
2. Any shocks between 2006-2009 that caused outages of GLOBEX for electronic traders, or would have prevented pit traders from reaching NYMEX?
3. It would be interesting to see more about the behavior in each category of account, do accounts vary their behavior and move from e-local to HFT and back again?

Conclusion

This paper makes an important contribution:

- ▶ Most empirical work on the impact of HFT needs to rely on DiD analysis
- ▶ But the introduction may not happen in these extreme periods
- ▶ And often we can't see if changes come from transactions between HFTs or Customer-HFT transactions (which is what we care about)
- ▶ The authors can distinguish between market-maker HFT and generic HFTs